

Designing a Glyph-Based Hierarchical Visualization for Orders of Magnitude Values

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How can we design a **hierarchical visualization** to support comparisons of **orders of magnitude values**, both within and across hierarchical levels?

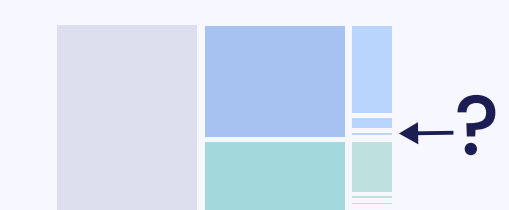
What are Orders of Magnitude Values (OMVs)?

Quantitative attributes that span over multiple orders of magnitude (e.g., from thousands to billions). Common in domains such as:

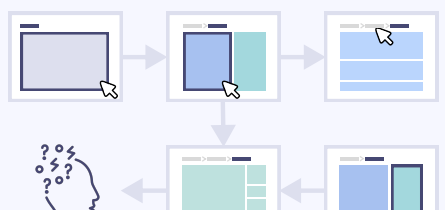


In hierarchical data, OMVs can occur within a level or across levels.

Can existing hierarchical visualizations effectively represent OMVs? - *Not really*

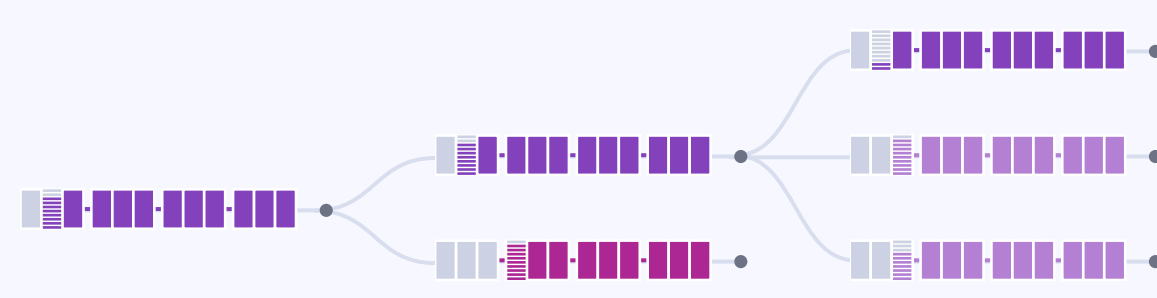


A. Common **space-filling** visualizations hide smaller magnitudes.



B. Zooming techniques result in **multi-view** visualizations, hindering comparisons across levels.

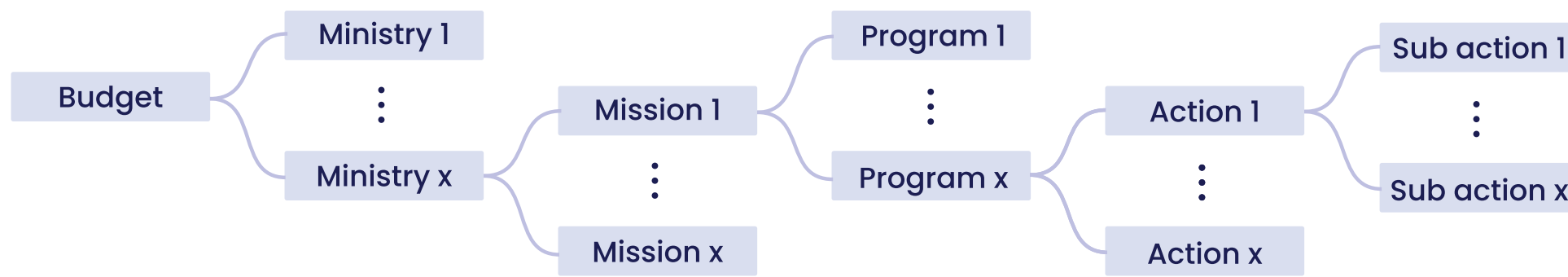
What do we propose?



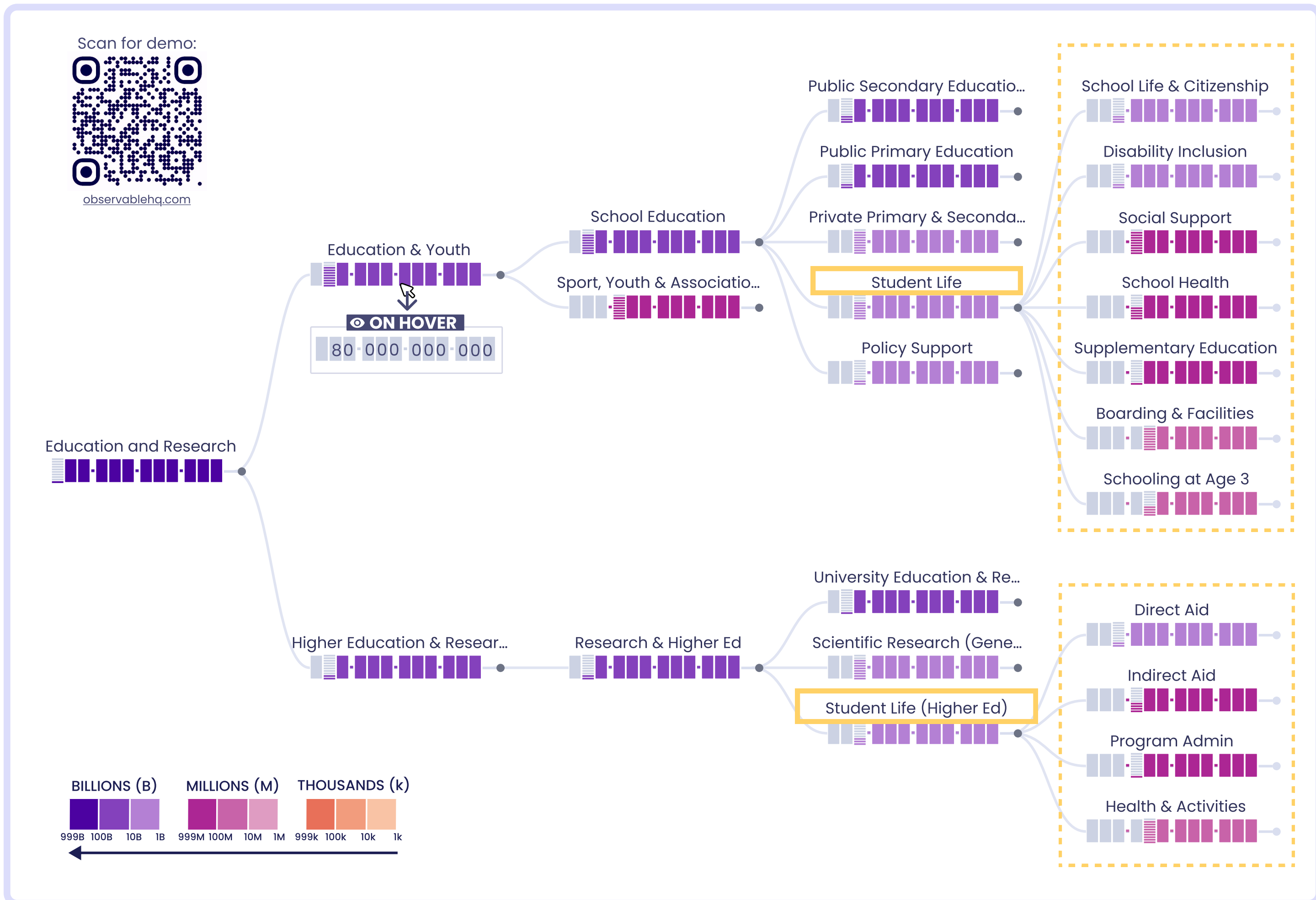
A **node-link layout** to support selective display of hierarchical levels.
A per-node **glyph** to support visual comparisons between OMVs.

DATA CASE: THE FRENCH NATIONAL BUDGET

The French national budget is hierarchically structured (Ministries → Missions → Programs → Actions → Sub Actions). Budgets across the hierarchy range from tens of thousands (10^4) to hundreds of billions (10^{11}), spanning eight orders of magnitude.

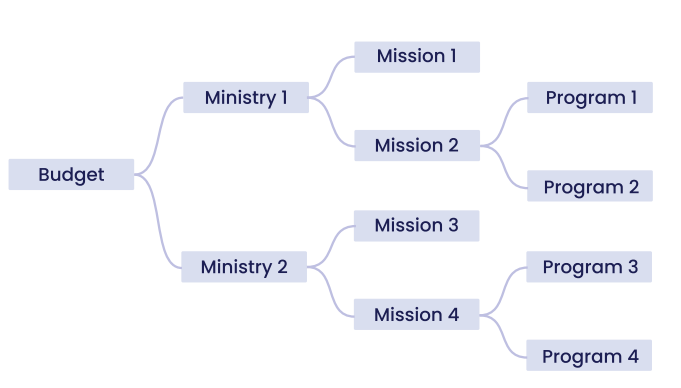


OUR PROPOSAL: A GLYPH-BASED NODE-LINK TREE



Node-Link Tree for Comparisons Across Hierarchies

✓ Selective Overview

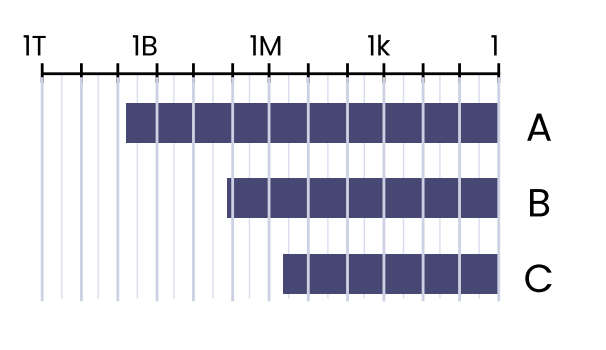


Widely used and familiar hierarchical visualization. Enables selective display of different levels through interaction.

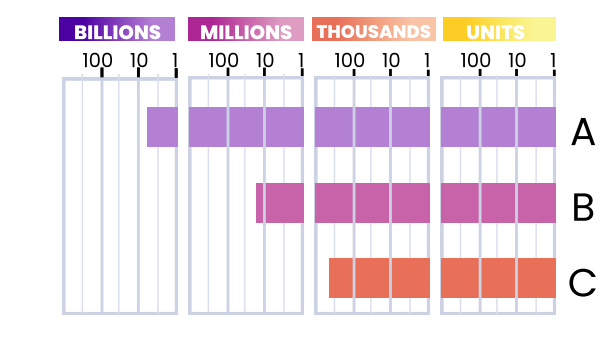
Glyphs for Orders of Magnitude Values

✓ Visibility of OMVs both across and within levels.

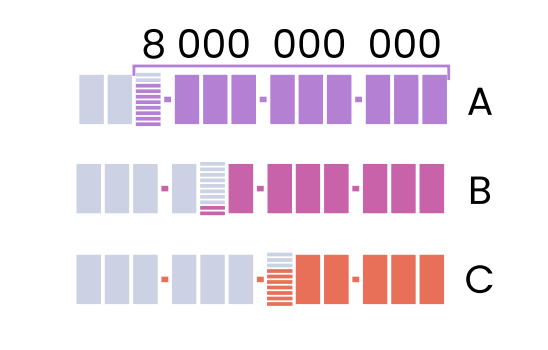
To reduce cognitive effort [4], we round values and decompose OMVs into three parts: (1) **mantissa**, (2) **exponent**, and (3) **scale words** (e.g., billions, millions) [1, 2, 3].



Encode OMVs using bars and the **EplusM** (Exponent plus Mantissa) scale [1]. Align right to reflect large-number reading.



Redundantly encode scale words with **faceting** and **colour hue**. Within each facet, decrease **colour intensity** per exponent.



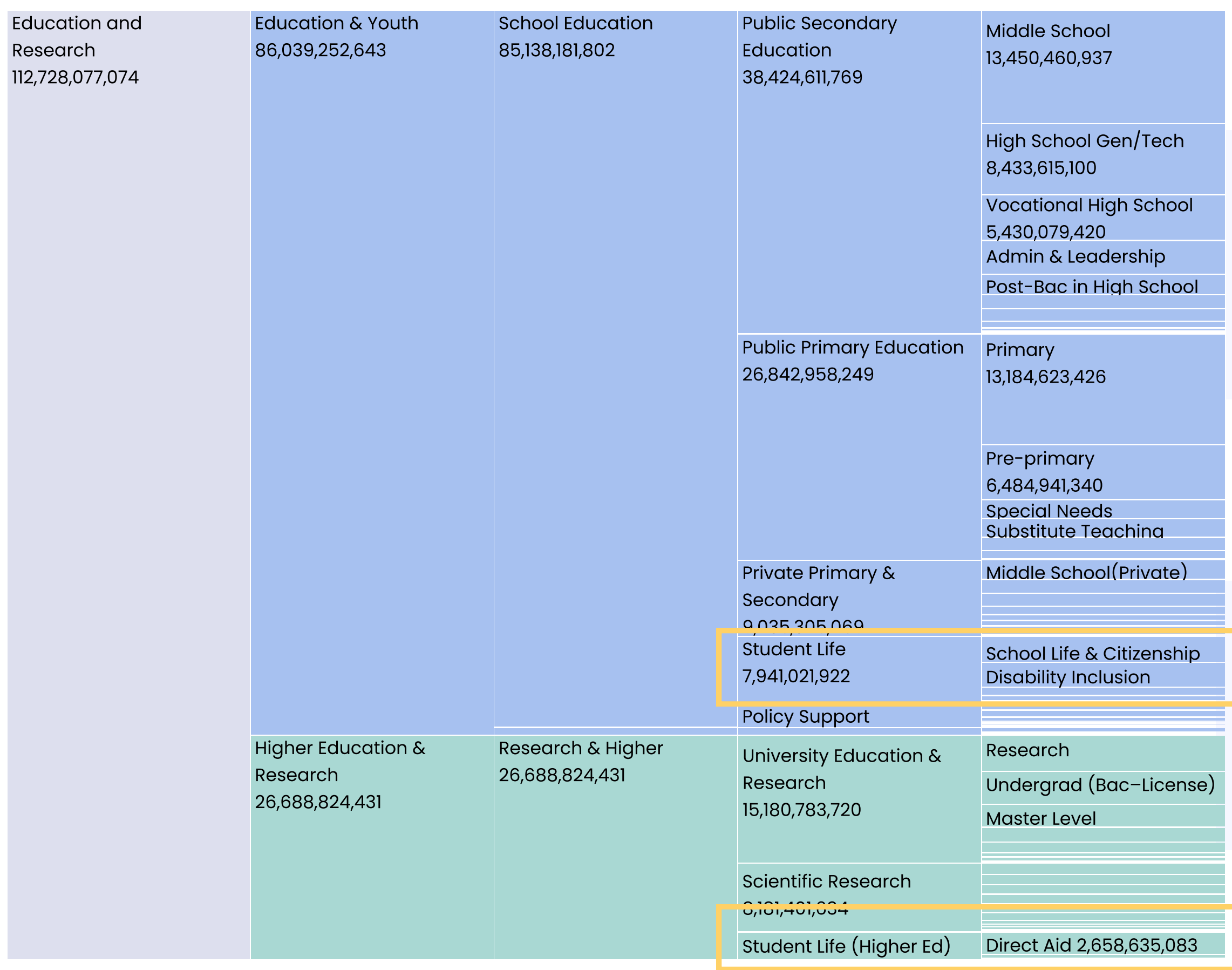
Isolate bars, each rectangle serves as a metaphor for a digit. Encode the mantissa with **discrete units** [1].

EXISTING HIERARCHICAL VISUALIZATIONS

A. Space-Feeling Hierarchical Visualizations

✓ Overview

✗ Do not support visibility of OMVs, either across or within levels.

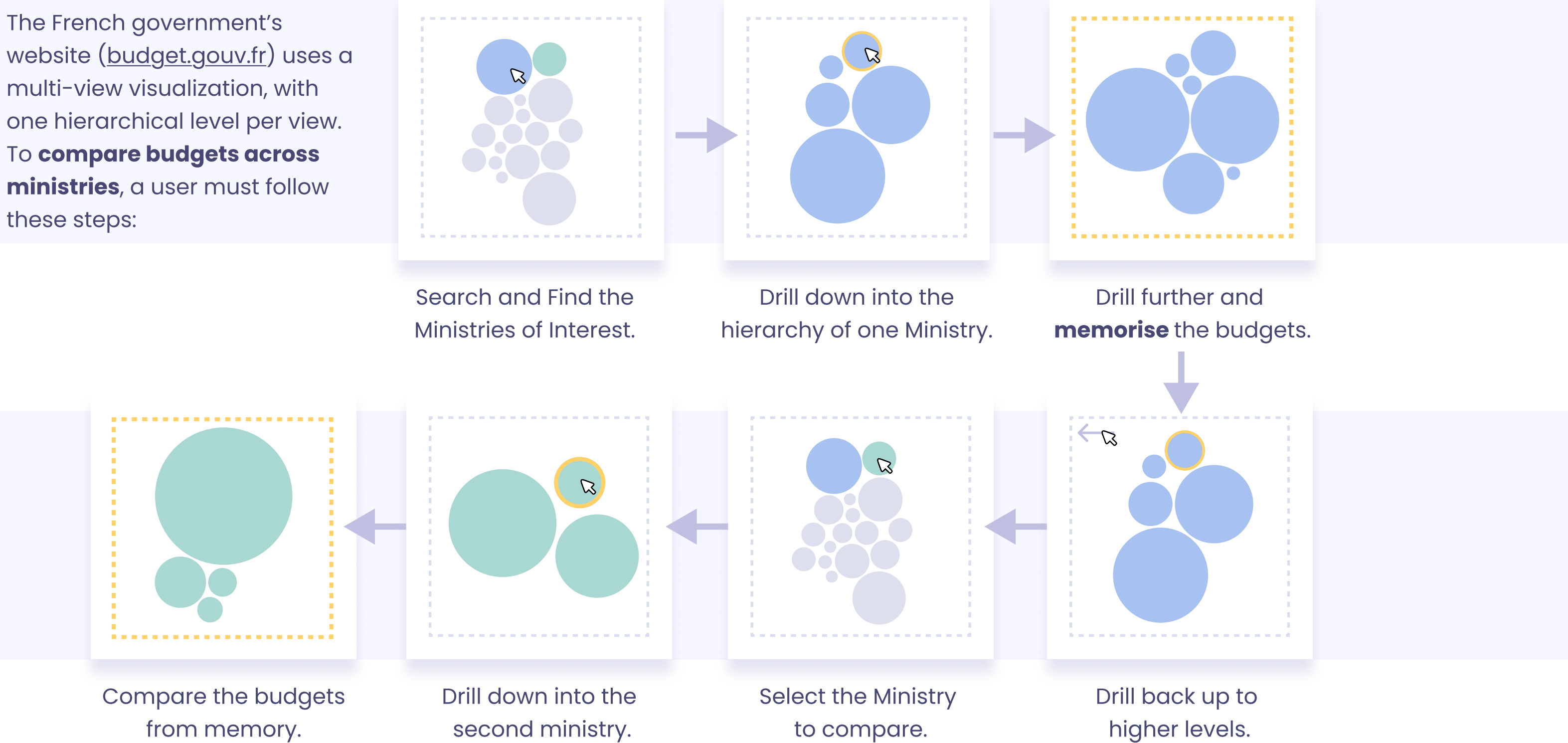


B. Multi-View Hierarchical Visualizations

✓ Better visibility of OMVs across levels, due to space reallocation.

✗ Do not support visibility of OMVs within a level.

✗ Users must remember previously seen nodes to compare, increasing cognitive load.



REFERENCES

- [1] K. Batziakoudi, S. Rey, and J.-D. Fekete. **Beyond Log Scales: Toward Cognitively Informed Bar Charts for Orders of Magnitude Values**. IEEE VIS 2025.
- [2] K. Batziakoudi, F. Cabric, S. Rey, and J.-D. Fekete. **Lost in magnitudes: Exploring visualization designs for large value ranges**. CHI '25. ACM, New York, NY, USA, 2025.
- [3] Landy, A. Charlesworth, and E. Ottmar. **Categories of Large Numbers in Line Estimation**. Cognitive Science, 41(2):326–353, 2017.
- [4] H. A. Nguyen, J. M. Hofman, and D. G. Goldstein. **Round numbers can sharpen cognition**. CHI '22. ACM, New York, NY, USA, 2022.